Khulna University of Engineering & Technology

Department of Urban and Regional Planning

BURP 1st Year 2nd Term Regular Examination, 2023

Hum 1217

(Principles of Economics)

Full Marks: 210 | Time: 3.00 hrs.

N.B. i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1 (a) What are the key problems of an economic organisation? Make a comparative study to solve the key problems through market and command economic systems. (15)

(b) Define opportunity cost. Explain economy's trade-off between high income and clean environment on a production possibilities frontier. (20)

2 (a) Suppose, the market demand and market supply equation for commodity x is given by the following equation, (35)

$$Q_d = 132 - 8p$$

$$Q_S = 6 + 4p$$

Now, answer the following questions:

i. Define law of demand.
ii. What is demand schedule and demand curve?
iii. Define price ceiling and floor price.
iv. Find out the equilibrium price and quantity.
v. What is the equilibrium price and quantity if unit tax K = 4.5?
vi. Explain the factors that affect the demand curve.

3 (a) What is meant by production in economics? Mention the functions of an entrepreneur in production process. (10)

(b) “A rational producer will always produce in stage-2” - critically explain the statement (20)

(c) Show the relationship between total cost, average cost and marginal cost. (05)

4 (a) Define market and illustrate the types of market. (05)

(b) What is perfect competition market? Show with the aid of a graph the short-run equilibrium of a firm in perfect competition. (20)

(c) Explain the shut-down position of a competitive firm with figure in the short-run. (10)

Section – B

5 (a) Define the concept Investment. Discuss the different categories of investment. (15)

(b) Why does full employment not stand for zero unemployment? Explain. (10)

(c) What is theory of distribution? Illustrate the concept of nominal wage and real wage. (10)

6 (a) What is inflation? What are the types of inflation? (07)

(b) Explain the causes of inflation. (08)

(c) To combat inflation, how do the government apply monetary instrument? Explain. (20)

7 (a) Define Pareto optimality. Analyze the conditions of pareto optimality. (15)

(b) Discuss the assumptions of input output model. (15)

(c) Solve the following equation for x. (05)

Equation: $X = AX + b$

8 (a) What is Commercial Bank? Interpret the primary and secondary functions of Commercial Bank. (15)

(b) Evaluate the difficulties of measuring national income. (20)

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Math 1217

(Mathematics - II)

Full Marks: 210 Time: 3.0 hrs

N.B. i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Define limit of a function at a point. Does (10)
$f(x) = \frac{x^2-4}{x-2}$ exist at $x = 2$?
If exists, find $\lim_{x\to 2} f(x)$.

(b) Discuss the continuity and differentiability of the function $f(x)$ at $x = 0$ (15)
Where $f(x) = \begin{cases} x\sin\left(\frac{1}{x}\right); x \neq 0 \\ 0; \quad x = 0 \end{cases}$

(c) Find the differential coefficient of y where $y = x^{\cos^{-1}x} + (\sin x)^{\ln x}$. (10)

2. (a) Write the statement of Leibnitz's theorem. If $y = \sin^{-1}x$, find the relations among y_{n+2}, y_{n+1}, y_n. (12)
(b) Find the maxima or minima of $\sin x\,(1 + \cos x)$. If exists, find the point of inflexion. (12)
(c) If $y = e^{ax}\sin bx$, find y_n. (11)

3. Integrate the followings:
(a) $\int \frac{dx}{(2x^2+5)\sqrt{x^2-4}}$ (12)
(b) $\int \frac{3\sin x-4\cos x-5}{\cos x-2\sin x+2}dx$ (12)
(c) $\int \frac{dx}{x^4+1}$ (11)

4. (a) Evaluate $\int_0^{\frac{\pi}{2}} \log\cos x\, dx$. (11)
(b) Evaluate $\int_0^{\pi} \frac{x}{a^2\sin^2x+ b^2\cos^2x} dx$ (12)
(c) Find the area of the segment cutoff from the parabola $y^2 = 4x$ by the straight line $y = 2x$. (12)

Section – B

5 (a) Define O.D.E. and P.D.E. with examples. From a differential equation from the relation $y = Ae^{-x} + Be^{2x} + x^2$ and indicate the order and degree of the obtained differential equation. (12)
(b) Solve $\frac{dy}{dx} = \frac{6x-2y-7}{3x-y+4}$. (11)
(c) Find a particular solution of $y'' - 2y' - 3y = 5$ where $y(0) = 2$ and $y'(0) = 4$. (12)

6. (a) Solve $(2x^2 + y)dx + (x^2y - x)dy = 0$. (13)
   (b) Solve $y^2dx + (3xy - 1)dy = 0$. (11)
   (c) Solve the initial value problem $2x(1 + y)dx + ydy = 0$ where $y(0) = -2$. (11)

7. (a) Solve $y^2dx + (3xy + y^2 - 1)dy = 0$. (11)
   (b) Solve $Cosy\frac{dy}{dx} + \frac{1}{x}siny = 1$. (12)
   (c) Solve $\frac{dy}{dx} = \sin(x + y) + \cos(x + y)$. (12)

8. (a) Solve $(D^2 - 3D + 2)y = sin3x; D = \frac{d}{dx}$. (11)
   (b) Solve $(D^2 + 4D + 4)y = x^3e^{-2x}$. (12)
   (c) Solve $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 2x^2 + cos2x$. (12)

Khulna University of Engineering & Technology

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BURP 1st Year 2nd Term Regular Examination, 2023

URP 1211

(Urban Planning Principles)

Full Marks: 210 Time: 3.00 hrs

N.B. i) Answer any three questions from each section in separate script.

ii) Figures in the right margin indicate full marks.

Section – A

1. (a) "Town planning is not God given or Natural, but has inextricable link with the prevailing economic system" – how so? Discuss in brief. (10)

   (b) "The proliferation of newest technologies has led to numerous challenges in satellite towns"- Discuss. (10)

   (c) Differentiate between the concept of Clarence A. Perry and Clarence Stein. (15)

2. (a) "Boulevards are monumental links between important destinations"- how so? Discuss. (15)

   (b) Do you believe that there is no distinction between a Satellite Town and a New Town? Give a concise explanation of your perspective. (10)

   (c) The land measurement is 58 feet * 65 feet. If FAR = 4 and MGC = 60%, find out the number of stories to be made for a residential building. (10)

3. (a) "Transit Oriented Development is a very much practical way of neighborhood planning rather than Radburn Concept"- do you agree? State your opinion with the necessary diagram. (15)

   (b) Suppose you are working as a town planner in Khulna Development Authority (KDA). Now, make a list of data that you are going to collect for planning and designing a bus terminal in Khulna City. (10)

   (c) "Railway stations play a vital role for representing the community"- Justify this statement using railway design principles. (10)

4. (a) "Open space is not vacant space"- why so? Discuss. (05)

   (b) "The principles of railway station and bus terminal is somewhat similar"- Discuss in brief. (20)

   (c) "Logistical access is the primary factor for choosing a best location for a seaport" - Discuss with relevant example. (10)

Section – B

5. (a) What is the rank-size rule in urban geography and how does it describe the population distribution of cities within a country? (10)

   (b) State the challenges associated with having a primate city in a country's urban hierarchy. How can urban planning strategies and policy interventions address these challenges? (15)

   (c) Explain why Central Place Theory uses hexagons for market areas and how this geometric choice enhances the theory's efficiency. (10)

6. (a) Discuss various types of towns based on their specific functions with necessary examples. (15)

   (b) Provide a comprehensive overview of the essential factors that one should take into account when determining the valuation of land. (15)

   (c) Illustrate the three types of radial pattern towns with necessary diagram. (05)

7. (a) Write a short note on the following (any four): (20)

   (i) Urban Periphery
   (ii) Urban Pyramid
   (iii) Collector Roads
   (iv) Arterials
   (v) Camber

   (b) Briefly discuss the relationships of functionally classified highway system in serving traffic mobility and land access. (15)

8. (a) "Industrial location within a town can be controlled or influenced by concession, persuasion and restriction" – Justify the statement. (08)

   (b) Present a case study illustrating the concept of industrial agglomeration, and subsequently, provide a concise analysis of its benefits and drawbacks. (15)

   (c) Narrate the parameters that one should consider for the development of an industrial park. (12)

Khulna University of Engineering & Technology

Department of Urban and Regional Planning

BURP 1st Year 2nd Term Regular Examination, 2023

URP 1251

(Introduction to Architectural History)

Full Marks: 210 Time: 3.0 hrs

N.B. i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) "The distinction between Architecture and Planning hinges on the scale of thought and execution involved in each discipline". Briefly explain the statement. (11)
   (b) "The climate of Bengal is indeed a decisive factor influencing the architecture of the region". Discuss the statement with necessary examples. (15)
   (c) Write down the short notes of (i) Timeless landscape (ii) Deltaic pavilion (iii) Vernacular settlement. (09)

2. (a) Briefly state the development of Buddhist Architecture in Bengal. (12)
   (b) Write down the philosophical interpretation of Vihara. (08)
   (c) Describe the architectural features of Kantaji Temple with necessary sketches. (15)

3. (a) "The architecture of Traditional Feudal Palace is the amalgam of Oriental Architecture and Occidental Architecture". Explain the notion. (20)
   (b) How did the development of Bangalow evolve from the Bengal Architecture? Explain. (15)

4. (a) What are the major architectural features of Mughal period? Explain with proper diagram. (15)
   (b) "Khan-e-Jahan style is the perfect example of how Organisationalism transform into individualism". Do you agree with the statement and why? (20)

Section – B

5. (a) What is architecture? How do you think lateral thought process and linear thought process co-exist in architecture? (15)
   (b) How can architecture be influenced? Between the two criteria "technology" and "human need", which one do you think pertinent as the most important one for influencing architecture? (20)

6. (a) Social and climate characteristics can shape the architecture of a civilization. Discuss it by referring to the architecture of Greece. Provide examples and illustrate if necessary. (20)
   (b) What is "Entasis" in Greek architecture? Describe it with the proper illustration. (15)

7. (a) What are the characteristics of Baroque movement? (10)
   (b) Describe the Renaissance and Baroque Portion of Saint's Peters Basilica. Provide necessary illustrations. (25)

8. (a) Describe any one of the Isms of Modern Architecture. (05)
   (b) Imagine, in a hypothetical situation, modern architecture didn't fail. How do you think our present architecture will then be impacted? Write both pros and cons. (20)
   (c) Which notion do you support-"less is more" or "less is bore"? Write your thought process behind it. (10)

Khulna University of Engineering & Technology

Department of Urban and Regional Planning

BURP 1st Year 2nd Term Examination, 2023

URP 1281

(Surveying and Cartography)

Full Marks: 210 Time: 3.00 hrs

N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Define the following with necessary sketch and example: (20)
   i) Well-conditioned and ill-conditioned triangles
   ii) Reconnaissance survey and index sketch
   iii) Criteria for selecting of survey stations
   iv) Large-scale and small-scale map

   (b) What is offset? Mention guidelines for number of offsets taken in the field for different object nature. Use necessary sketch. (15)

2. (a) Define the following with necessary sketch and example: (20)
   i) True meridian and magnetic meridian
   ii) Whole circle bearing and reduced bearing
   iii) Fore and back bearing
   iv) Magnetic declination

   (b) What is closing error in a traverse? Describe, with a sketch, how such an error is adjusted. (15)

3. (a) What are the methods of plane tabling? Describe them with necessary sketch. (15)

   (b) The following consecutive readings were taken with a level and a 4-meter leveling staff on a continuously sloping ground at common intervals of 50 m: (20)
   3.776 m (on A), 2.502 m, 2.323 m, 3.128 m, 3.813 m, 4.056 m, 4.367 m, 3.056 m, 3.434 m, 1.518 m, 0.587 m, 3.705 m, 4.507 m and 4.364 m (on B).
   The RL of A was 137.500 m. Make entries in a level book and apply the usual checks. The change points were taken at 150 m and 400 m in chainage.

4. (a) What is a contour line? Describe the characteristics of contour lines, including relevant sketches. (15)

   (b) What are the common applications of GPS, and how are they utilized across various fields? Provide examples to illustrate these uses. (10)

   (c) What are the advantages and disadvantages of using a total station in surveying? (10)

Section – B

5. (a) "Without maps we are spatially blind" – State your argument in favor or against this statement. (10)

   (b) What are the types and classification of maps? Explain with example and Drawings. (25)

6. (a) Why is BTM projection system used in Bangladesh instead of having global projection system (UTM)? (10)

   (b) Describe the Universal Transverse Mercator (UTM) Coordinate System with respect to Bangladesh. (10)

   (c) What do you mean by large scale and small-scale maps? Two maps are drawn at scales of 1:1717017 and 1:2217017 respectively. Which of the two maps will show a greater degree of detail and why? (15)

7. (a) An area of 17 sq. inches in drawing represents an area of 1717 sq.km. What is the R.F. of the scale? Also, define R.F. in your own words. (15)

   (b) Do you think "Spatial Reference = Datum + Projection + Coordinate system"? State your argument in favor or against this statement. (20)

8. (a) Describe general principles and elements of making better maps. (15)

   (b) Write short note on – (10)
   Georeferencing and Spatial Adjustment

   (c) "Distortions in maps make geographers S.A.D.D."- Explain the statement with relevant examples of various types of distortions. (10)